

NAME : _____
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QUESTION 1

Compile the given codes and study its output.

```
import java.util.*;

public class BubbleSortExample
{
    static void bubbleSort(int[] arr)
    {
        int n = arr.length;
        int temp = 0;

        for(int i = 0; i < n; i++)
        {
            for(int j=1; j < n; j++)
            {
                if(arr[j-1] > arr[j])
                {
                    temp = arr[j-1];
                    arr[j-1] = arr[j];
                    arr[j] = temp;
                }
            }
        }
    }

    public static void main(String[] args)
    {
        int arr[] = {3, 60, 35, 2, 45, 320, 5};

        System.out.println("Original contents of the array");
        for(int i=0; i < arr.length; i++)
            System.out.print(arr[i] + " ");

        System.out.println();

        bubbleSort(arr);

        System.out.println("Contents of the array after sorting");
        for(int i=0; i < arr.length; i++)
            System.out.print(arr[i] + " ");
    }
}
```

QUESTION 2

Given the following ADT:

```
public class PreSchool
{
    private String name;
    private String race; //Malay, Chinese, Indian
    private int age;

    //constructors
    //getters
    //printer
}
```

Write program segments to do the following tasks:

- (a) Create an `ArrayList` object named `PSList` to store data of 5 children.
- (b) Input the data of FIVE (5) children into the `PSList`.
- (c) Count and display the number of children for each race.
- (d) Given that every 4 years old preschooler is charged RM400 and 5 years old and 6 years old children are charged RM450, count and display the total collection for the preschool at the end of every month.

QUESTION 3

Based on the given codes, write the method definition for `bubbleSort()`.

```
import java.util.*;

public class BubbleSortList1
{
    public static void main(String[] args)
    {
        PreSchool ps = new PreSchool();
        ArrayList PSList = new ArrayList();

        ps = new PreSchool("suffian", "malay", 5);
        PSList.add(ps);
        ps = new PreSchool("alma", "malay", 6);
        PSList.add(ps);
        ps = new PreSchool("jenny", "chinese", 3);
        PSList.add(ps);
        ps = new PreSchool("gopal", "indian", 4);
        PSList.add(ps);
        ps = new PreSchool("mei mei", "chinese", 3);
        PSList.add(ps);

        System.out.println("Original contents of the array");
        System.out.print( PSList + " ");
        System.out.println();

        bubbleSort(PSList);
    }
}
```

```
        System.out.println("\n\nContents of the array after sorting");  
        System.out.print( PSList + " ");  
        System.out.println();  
    }  
}
```

QUESTION 4

Given the following ADT:

```
public class Car  
{  
    private String registrationNo;  
    private String ownerName;  
    private String manufacturerName;  
    private double carPrice;  
  
    //constructors  
    //getters  
    //printer  
}
```

By assuming that several Car objects has been inserted into an `ArrayList` named `carRecords`, write a program segment to:

- (a) Find and display the information of a car based on the `registrationNo` given by the user. If the information does not exist, display appropriate message.
- (b) Count and display the number of cars registered in Perlis.
- (c) Display the name of the owner with the most expensive car.
- (d) Find and remove all the “Honda” cars.
- (e) Display the current size of `carRecords`.